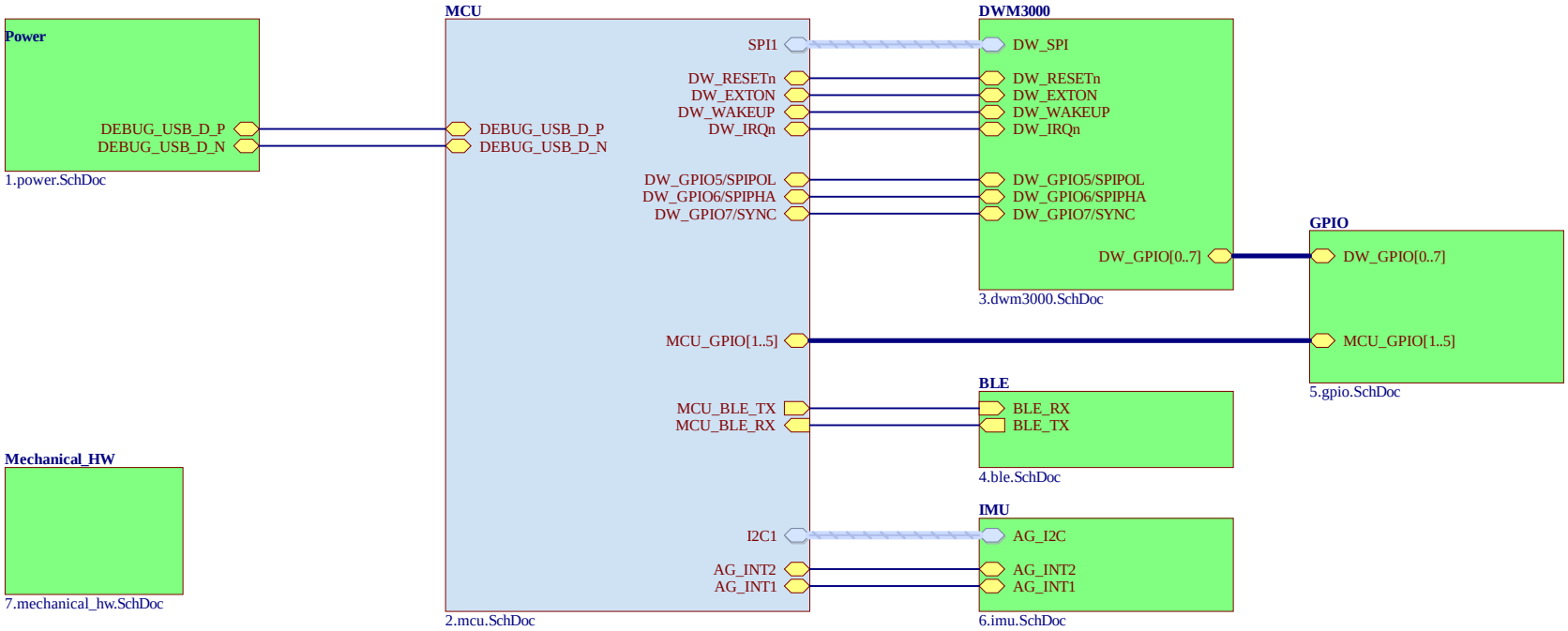


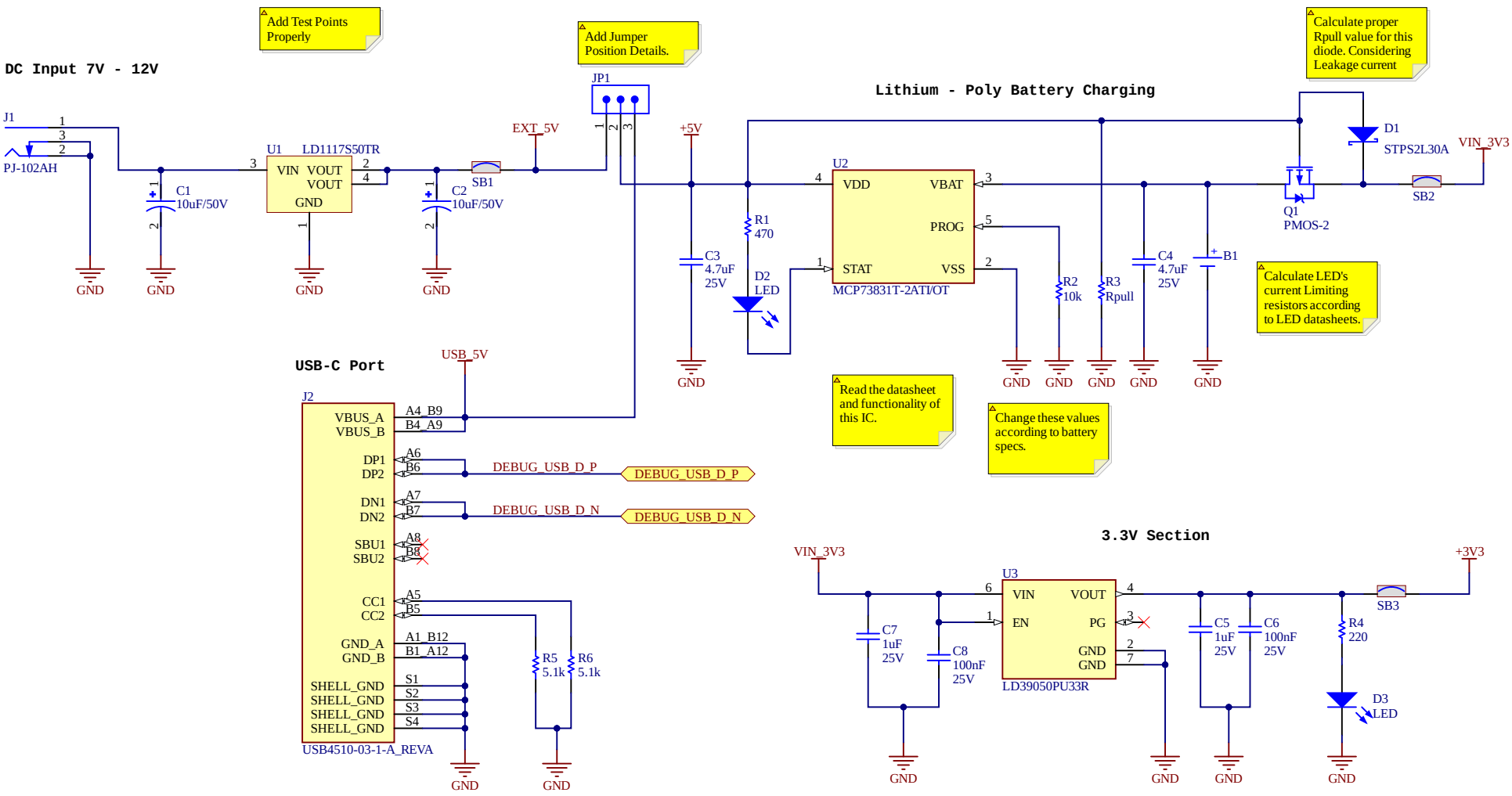
Schematic / General Notes

Caution / Warning Notes

PCB Layout Notes

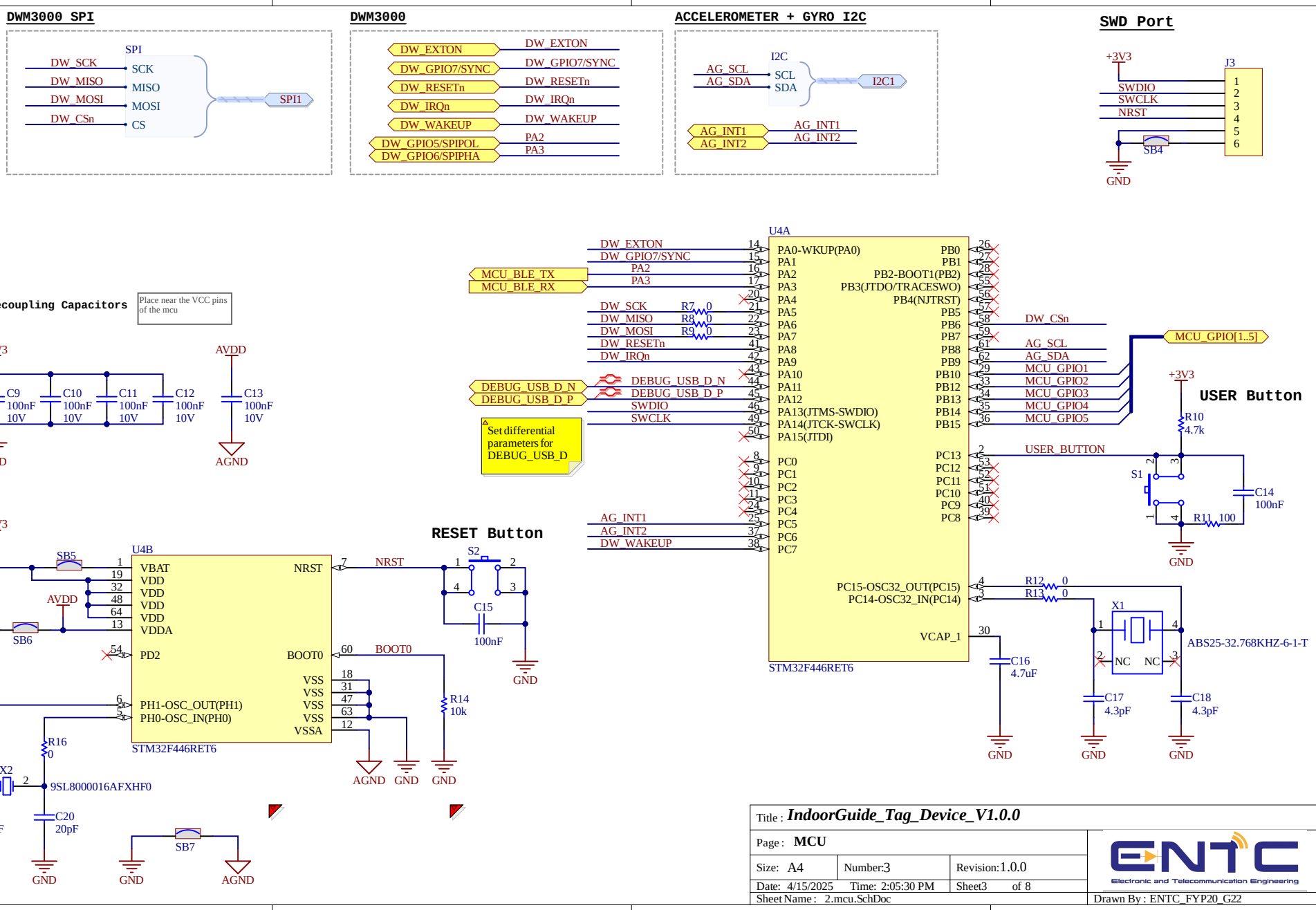
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Page : **Top**		
Size: A4	Number:1	Revision:1.0.0
Date: 4/15/2025 Time: 2:05:30 PM	Sheet1 of 8	
Sheet Name : 0.main.SchDoc		Drawn By : ENTC_FYP20_G22

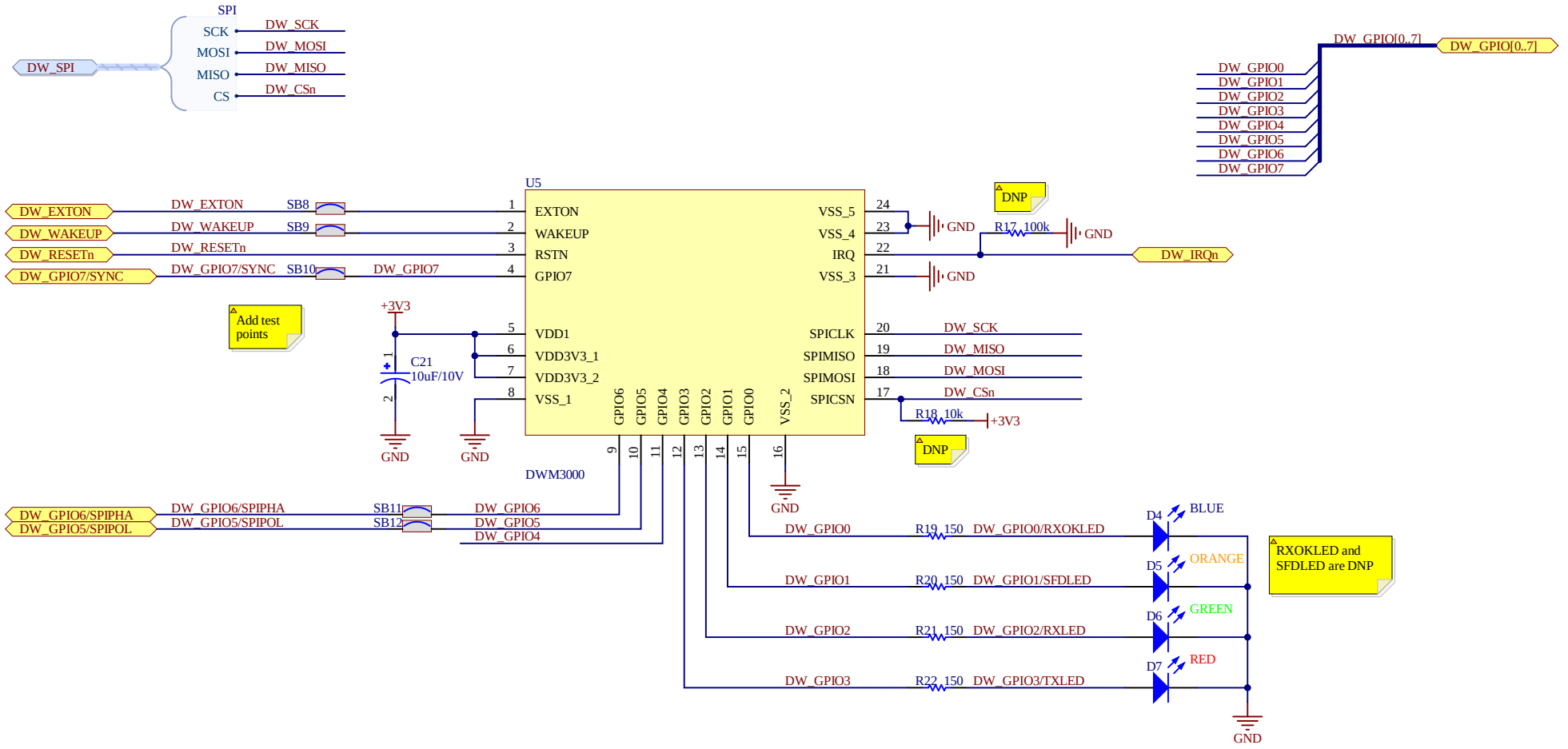
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Page : **Power**		
Size: A4	Number:2	Revision:1.0.0
Date: 4/15/2025 Time: 2:05:30 PM	Sheet2 of 8	
Sheet Name : 1.power.SchDoc		Drawn By : ENTC_FYP20_G22

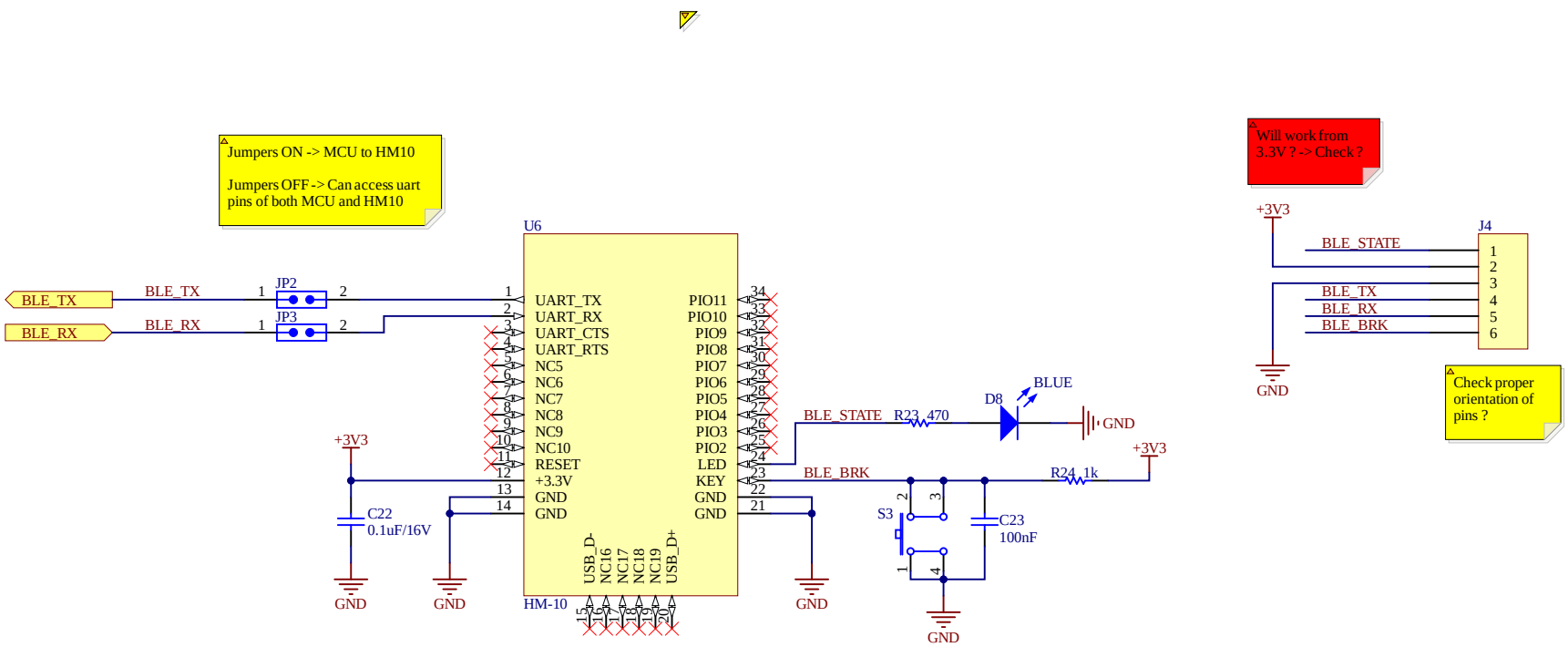
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DWM3000 SPI
SPI
DW_SCK
DW_MISO
DW_MOSI
DW_CSn
SCK
MISO
MOSI
CS
SPI1
DWM3000
DW_EXTON
DW_GPIO7/SYNC
DW_RESETn
DW_IRQn
DW_WAKEUP
DW_GPIO5/SPIPOL
DW_GPIO6/SPIPHA
PA2
PA3
ACCELEROMETER + GYRO I2C
I2C
AG_SCL
AG_SDA
SCL
SDA
I2C1
AG_INT1
AG_INT2
SWD Port
+3V3
J3
SWDIO
SWCLK
NRST
SB4
GND
Decoupling Capacitors
Place near the VCC pins of the mcu
C9 100nF 10V
C10 100nF 10V
C11 100nF 10V
C12 100nF 10V
C13 100nF 10V
AVDD
AGND
U4A
PA0-WKUP(PA0)
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC13
PC12
PC11
PC10
PC9
PC8
PC15-OSC32_OUT(PC15)
PC14-OSC32_IN(PC14)
VCAP_1
STM32F446RET6
MCU_BLE_TX
MCU_BLE_RX
DEBUG_USB_D_N
DEBUG_USB_D_P
SWDIO
SWCLK
R7 0
R8 0
R9 0
Set differential parameters for DEBUG_USB_D
MCU_GPIO[1..5]
MCU_GPIO1
MCU_GPIO2
MCU_GPIO3
MCU_GPIO4
MCU_GPIO5
USER_BUTTON
USER Button
R10 4.7k
S1
C14 100nF
R11 100
RESET Button
S2
C15 100nF
NRST
BOOT0
R14 10k
U4B
VBAT
VDD
VDDA
PD2
PH1-OSC_OUT(PH1)
PH0-OSC_IN(PH0)
VSS
VSSA
STM32F446RET6
SB5
SB6
SB7
R15 0
R16 0
X2
9SL8000016AFXHF0
C19 20pF
C20 20pF
R12 0
R13 0
X1
ABS25-32.768KHZ-6-1-T
NC
C16 4.7uF
C17 4.3pF
C18 4.3pF
Title : IndoorGuide_Tag_Device_V1.0.0
Page : MCU
Size: A4
Number:3
Revision:1.0.0
Date: 4/15/2025 Time: 2:05:30 PM
Sheet3 of 8
Sheet Name : 2.mcu.SchDoc
Drawn By : ENTC_FYP20_G22
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Title : ***IndoorGuide_Tag_Device_V1.0.0***			
Page : **DW3000**			
Size: A4	Number:4	Revision:1.0.0	
Date: 4/15/2025	Time: 2:05:31 PM	Sheet4 of 8	
Sheet Name : 3.dwm3000.SchDoc			Drawn By : ENTC_FYP20_G22

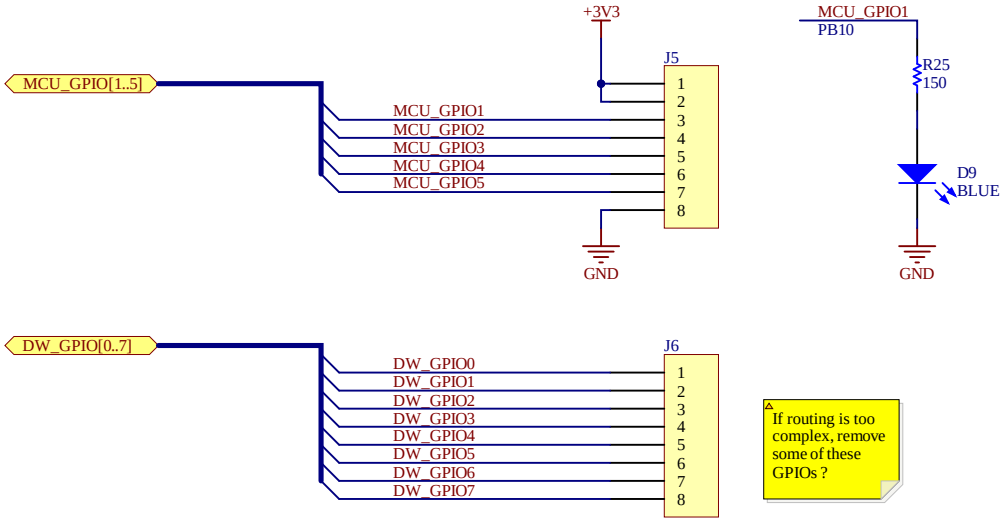
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Title : ***IndoorGuide_Tag_Device_V1.0.0***		
Page : **Ble_module**		
Size: A4	Number:5	Revision:1.0.0
Date: 4/15/2025 Time: 2:05:31 PM	Sheet5 of 8	
Sheet Name : 4.ble.SchDoc		Drawn By : ENTC_FYP20_G22

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USER LEDs

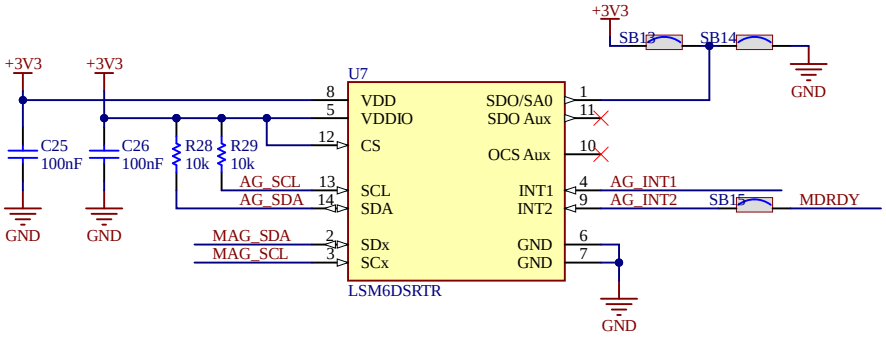


Title : ***IndoorGuide_Tag_Device_V1.0.0***			
Page : **GPIO**			
Size: A4	Number:6	Revision:1.0.0	
Date: 4/15/2025 Time: 2:05:31 PM		Sheet6 of 8	
Sheet Name : 5.gpio.SchDoc			Drawn By : ENTC_FYP20_G22

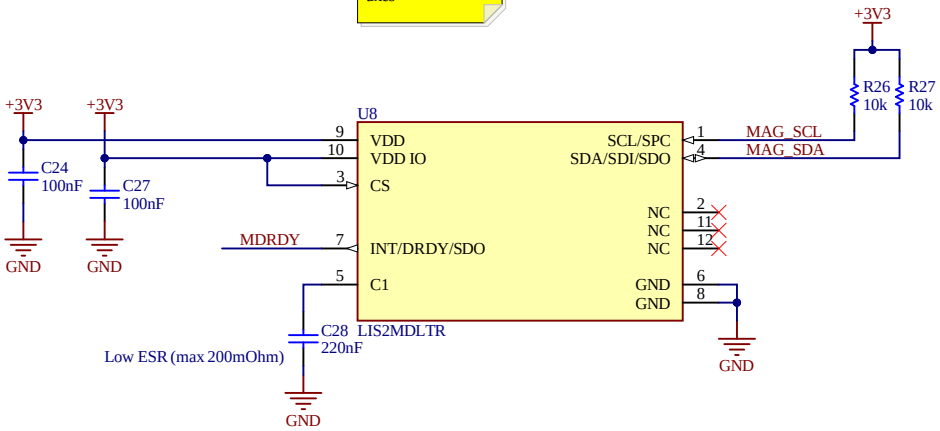
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Check Footprints properly.



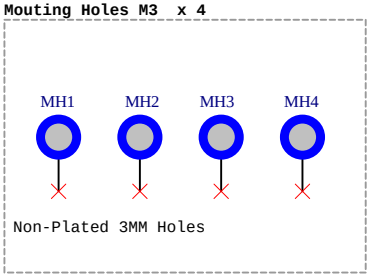
Properly place IMU ics X,Y,Z axes



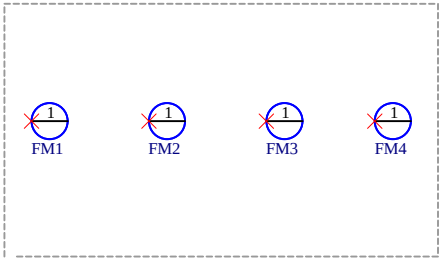
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Page : **IMU**			
Size: A4	Number:7	Revision:1.0.0	
Date: 4/15/2025	Time: 2:05:31 PM	Sheet7 of 8	
Sheet Name : 6.imu.SchDoc			Drawn By : ENTC_FYP20_G22

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Mouting Holes M3 x 4



Fiducial Marks x 4



Title : ***IndoorGuide_Tag_Device_V1.0.0***			
Page : **Mechanical**			ENTC Electronic and Telecommunication Engineering
Size: A4	Number:8	Revision:1.0.0	
Date: 4/15/2025 Time: 2:05:31 PM		Sheet8 of 8	
Sheet Name : 7.mechanical_hw.SchDoc			Drawn By : ENTC_FYP20_G22